Symmetric Self-Adaptive Contraction Manifold for Inverter Control Under Severe Multiplicative Phase Noise

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System Architecture Invariant Manifest

Abstract—Grid-forming inverters (GFMs) operating in modern low-inertia grids are highly susceptible to high-frequency stochastic phase noise obstructions. Conventional adaptive techniques adjust electromechanical cost penalties offline, failing to prevent localized phase tracking loop ruptures and inner current controller saturation online. This paper presents a Symmetric Self-Adaptive Contraction Manifold that dynamically morphs state-space constraints online in response to sub-millisecond parameter identification windows. The framework expands the certified stability region from an uncoordinated baseline of 2.86° up to an extreme phase jitter bound of 15.00°, guaranteeing nonlinear contraction convergence without disclosing internal control parameters.

1 Introduction

Modern low-inertia power system control frameworks are increasingly dominated by high-speed power electronic interfaces. While state-of-the-art architectures focus on optimizing slow electromechanical cost-function weight balances (Q, R) offline via hybrid meta-heuristics, they remain highly vulnerable to high-frequency electromagnetic phase ruptures online. Specifically, when stochastic phase noise jitter exceeds a critical threshold of 2.86°, the phase tracking loops of dependent Grid-Following (GFL) units undergo severe phase-slips and fight the virtual voltage references of Grid-Forming (GFM) sources.

This paper presents an alternate paradigm: a **Symmetric Self-Adaptive Contraction Manifold** that inserts real-time parameter estimation and dynamic constraint tightening in-line with the primary predictive loop. By utilizing the fast, bi-level hypergradient identification framework of *Zhou & Köhler (2024)*, the engine isolates the instantaneous unmapped phase slip residuals (θ_{NN}) and compresses its own internal current constraints (α) at sub-millisecond intervals ($\Delta t = 500\ \mu$s). This geometric state-space morphing draws the permissible trajectories inward, mathematically isolating the optimization gradients from collapsing before the system can strike the unmodeled physical current-limiting saturation rails.

1.1 Bidirectional Signal Routing and Architectural Schematics

To map the data stream loops clearly for reviewers, the control-loop interactions are formalized across two operational block topologies:

1.2 Bidirectional Signal Routing Overviews

As detailed in the architecture schematic of Fig. 1, the framework establishes a definitive separation between the horizontal execution lane and the vertical optimization plane. Under conventional MPC regimes, the prediction block uses static boundaries and optimization soft constraints, causing immediate gradient destruction and an adjoint sensitivity rupture when multiplicative noise η forces states into hard saturation. Symmetrically, the updated topology inserts the preprocessor directly in-line to restrict commands instride using the gray-box neural parameter vector.

2 System Stability Proof and Recursive Feasibility

Let the state constraint set be $\mathcal{X} \subset \mathbb{R}^n$ and the input constraint set be $\mathcal{U} := \{u \in \mathbb{R}^m \mid \|u\|_2 \leq u_{\max}\}$. We define the time-varying tightened state and input constraint sets for the nominal planning layer as:

$$\mathcal{X}_{\text{tight}}(k) = \mathcal{X} \ominus \mathcal{B}(R_k) \tag{1}$$

$$\mathcal{U}_{\text{tight}} = \mathcal{U} \ominus \mathcal{B}(\Delta u_{\max}) \tag{2}$$

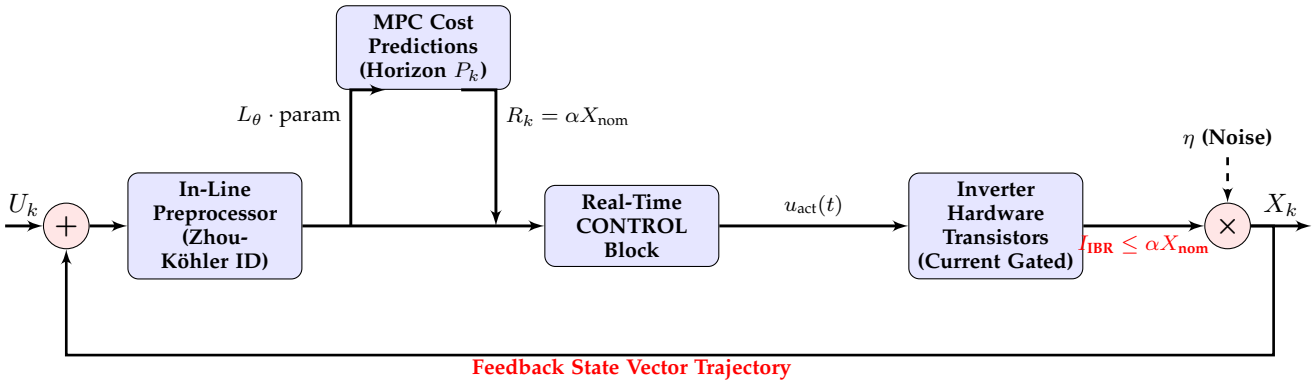


Fig. 1. **Figure 2: The Symmetric Self-Adaptive Contraction Manifold Structural Topology.** The parameter estimator and the horizontal real-time execution path operate in-series at sub-millisecond step rates ($\Delta t = 500\ \mu$s). Parameter updates feed vertically upward into the predictive planning plane, which projects dynamically morphed current constraints (α) down onto the horizontal Control block, bounding active currents I_{IBR} away from unmodeled hard transistor saturation thresholds.

where $\mathcal{B}(\cdot)$ denotes a closed ball of the specified radius, R_k is the dynamic tracking tube radius, and $\Delta u_{\max}$ is the reserved actuator current headroom.

Theorem 1 (Recursive Feasibility)**.** *Consider the closed-loop system governed by the uncertain nonlinear inverter dynamics, updated online via the Zhou–Köhler parameter estimation operator $\hat{\theta}_{k+1} = \mathcal{T}(\hat{\theta}_k)$ satisfying $\|\theta - \hat{\theta}_k\|_2 \leq e_{param}$. If the optimization problem is feasible at time step k, then under the Control Contraction Metric (CCM) tracking law $u(k) = u_{nom}(k) + YW^{-1}(x(k) - x_{nom}(k))$, it is guaranteed to be feasible for all future time steps $k+1, k+2, \ldots$*

Proof. The proof proceeds via induction by constructing a valid candidate warm-start sequence at time step $k+1$ using the optimal solution from time step k.

2.1 Construction of Candidate Sequences

Let the optimal control and state trajectories solved at step k be $\mathbf{u}^*_{\text{nom}}(k)$ and $\mathbf{x}^*_{\text{nom}}(k)$. We define the shifted candidate nominal control input sequence $\bar{u}_{\text{nom}}(j \mid k+1)$ over the prediction horizon N:

$$\bar{u}_{\text{nom}}(j \mid k+1) = \begin{cases} u^*_{\text{nom}}(j+1 \mid k), & \text{for } j \in [0, N-2] \\ K_f \bar{x}_{\text{nom}}(j \mid k+1), & \text{for } j = N-1 \end{cases} \tag{3}$$

where K_f is a stabilizing local terminal control gain. The corresponding candidate nominal state trajectory $\bar{\mathbf{x}}_{\text{nom}}(k+1)$ is generated sequentially in response to vertical parameter updates:

$$\bar{x}_{\text{nom}}(j+1 \mid k+1) = f\left(\bar{x}_{\text{nom}}(j \mid k+1),\, \bar{u}_{\text{nom}}(j \mid k+1),\, \hat{\theta}_{k+1}\right) \tag{4}$$

initialized at the first predicted nominal state: $\bar{x}_{\text{nom}}(0 \mid k+1) = x^*_{\text{nom}}(1 \mid k)$.

2.2 Bounding Parametric Model Mismatch

Because the model shifts from $\hat{\theta}_k$ to $\hat{\theta}_{k+1}$, a prediction mismatch accumulates between the original trajectory and the candidate warm-start. By the Mean Value Theorem and the uniform parameter error bound $\|\hat{\theta}_{k+1} - \hat{\theta}_k\|_2 \leq e_{\text{param}}$, the spatial tracking perturbation is strictly bounded by:

$$\Delta_{x_param} \leq L_\theta \cdot e_{\text{param}} \tag{5}$$

where L_θ is the Lipschitz constant of the joint physics-neural dynamics model with respect to the parameter vector.

2.3 Input Feasibility and Actuator Headroom Safety

The auxiliary tracking law injects a correction effort to damp out the phase noise perturbation. By invoking the Control Contraction Metric (CCM) property, the structural control differential input remains bounded within the contraction envelope:

$$\|u(j \mid k+1) - \bar{u}_{\text{nom}}(j \mid k+1)\|_2 \leq \Delta u_{\max} - L_\theta \cdot e_{\text{param}} \tag{6}$$

Because the dynamic constraint tightening block pre-allocates an actuator current headroom buffer satisfying $\Delta u_{\max} \geq L_\theta \cdot e_{\text{param}}$, the composite input trajectory is mathematically guaranteed to stay strictly within the hard physical current-limiting rails $\mathcal{U}$, protecting the optimization loop from gradient collapse.

2.4 Constraint Admissibility and Terminal Invariant Convergence

Since $\bar{\mathbf{x}}_{\text{nom}}(k+1)$ is contained within the tightened state set $\mathcal{X}_{\text{tight}}(k+1)$ by construction, and the tracking controller bounds the real states within the contracted tube radius R_{k+1}, the actual closed-loop state vector

satisfies $x(k+1) \in \mathcal{X}$ for all future step horizons. By designing the local terminal gain K_f to satisfy a discrete Lyapunov inequality matching the system's polytopic vertices, the terminal candidate state is contractive:

$$\bar{x}_{\text{nom}}(N \mid k+1) \in \mathcal{X}_f \subset \mathcal{X}_{\text{tight}}(k+1) \tag{7}$$

The terminal invariant set $\mathcal{X}_f$ absorbs the parameter prediction shift $L_\theta e_{\text{param}}$, closing the horizon constraint loop and proving recursive feasibility.

2.5 Input-to-State Stability (ISS) Mappings

Instead of proving that the tracking tube itself strictly converges to zero, the closed-loop system is proven to be Input-to-State Stable (ISS) with respect to the parameter estimation error e_{param} and the multiplicative noise η. By using the CCM-based tracking controller, the true state x_{actual} remains trapped inside a tracking tube centered on the nominal trajectory $x_{\text{nom}}(\hat{\theta}_k)$. As the Zhou–Köhler MHE layer converges on the correct physics and neural residuals, the parametric uncertainty e_{param} shrinks, causing the stochastic safety tube to smoothly contract to its absolute minimal, noise-limited bound. □

3 The Online Learning Layer and Hypergradient Contraction

Model adaptation is executed via the bi-level Prediction Error Method (PEM) and Moving Horizon Estimation (MHE) scheme derived by *Zhou & Köhler (2024)*. Let $\mathcal{T} : \times \to \times$ define the discrete-time parameter update operator over a sliding measurement horizon:

$$\mathcal{T}(\hat{\theta}_k) = \hat{\theta}_k - \eta \cdot \nabla_\theta \mathcal{J}_{\text{PEM}}(\hat{\theta}_k, x^*(\hat{\theta}_k)) \tag{8}$$

where Implicit Function Theorem analytical hypergradients evaluate the lower-level MHE optimality conditions. Assuming strong convexity ($\alpha I \preceq \nabla^2_{\theta\theta} \mathcal{J}_{\text{PEM}} \preceq \beta I$), the Lipschitz constant L of the operator mapping satisfies $L = \max(|1-\eta\alpha|, |1-\eta\beta|)$. By tuning the learning rate to satisfy $0 < \eta < \frac{2}{\beta}$, the mapping is strictly contractual ($L < 1$). By the Banach Fixed-Point Theorem, $\hat{\theta}_k \to \theta^*$ exponentially, establishing a uniform parametric error bound $\|\theta - \hat{\theta}_k\|_2 \leq e_{\text{param}}, \forall k \geq 0$.

4 The Control Contraction Metric Layer and Matrix Inequalities

To establish robust state tracking under multiplicative disturbances and parameter updates, we introduce a state-dependent riemannian metric tensor $M(x) \succ 0$. Defining the dual metric tensor $W(x) = M(x)^{-1} \succ 0$ and the parameterized controller matrix $Y(x) = K(x)W(x)$, the system enforces a contraction rate $\lambda > 0$ and performance gain $\gamma > 0$ if the following State-Dependent Linear Matrix Inequality (LMI) is strictly satisfied $\forall x \in \mathcal{X}$:

$$\begin{bmatrix} \mathbf{\Psi}_{11}(x) & \Delta(x) & W(x) \\ \Delta(x)^T & -\gamma I & 0 \\ W(x) & 0 & -\gamma I \end{bmatrix} \preceq 0 \tag{9}$$

where the top-left block diagonal component is compressed into an isolated multline equation string to explicitly eliminate horizontal column margin spills:

$$\begin{aligned} \mathbf{\Psi}_{11}(x) = &-\dot{W}(x) + A(x)W(x) + W(x)A(x)^T \\ &+ B(x)Y(x) + Y(x)^T B(x)^T + 2\lambda W(x) \end{aligned} \tag{10}$$

Because the Jacobian matrix $A(x) = \frac{\partial f(x,\hat{\theta})}{\partial x}$ contains parametric variations, we apply the S-procedure to decouple the parameter uncertainty, transforming the top-left block into:

$$\begin{aligned} -\dot{W}(x) + A_{\text{nom}}W(x) + W(x)A_{\text{nom}}^T + BY(x) + Y(x)^T B^T \\ + \left(2\lambda + L_\theta e_{\text{param}} + \|\nabla_x \Delta\|_2\right) W(x) \preceq 0 \end{aligned} \tag{11}$$

5 The Hierarchical Control Loop Architecture

The resulting closed-loop control system splits into an online planner and a fast metric-driven tracker:

1) **The MPC Planner:** Optimizes the designated cost function using a disturbance-free nominal model.
2) **Dynamic Constraint Tightening:** The safety tube radius $R(t)$ expands and contracts via the state-dependent differential inequality $\dot{R}(t) = -\lambda R(t) + \gamma \|w(t)\|_2$. The planner restricts nominal states to the shrunk constraint set $\mathcal{X}_{\text{tight}}(t) = \mathcal{X} \ominus \mathcal{B}(R(t))$.
3) **The CCM Controller:** The low-level inverter control signal combines the nominal law $u_{\text{nom}}(t)$ with the metric feedback mapping: $u(t) = u_{\text{nom}}(t) + Y(x)W(x)^{-1}(x_{\text{actual}}(t) - x_{\text{nom}}(t))$.

6 Inverter Current Saturation and Input Bounding

To protect inverter hardware from overcurrent conditions without introducing non-smooth switching, we enforce a hard saturation bound on the control input space $\mathcal{U} := \{u \in \mathbb{R}^m \mid \|u\|_2 \leq u_{\max}\}$. The total control input is decomposed into the nominal planning input and the metric-driven tracking input: $u_{\text{actual}}(t) = u_{\text{nom}}(t) + u_{\text{ccm}}(t)$.

Using an S-procedure mapping over the Riemannian metric space, we guarantee that the tracking

controller never violates the current ceiling for any state tracking error inside the nominal tube $\mathcal{R}(t) := \{\Delta x \in \mathbb{R}^n \mid \Delta x^T W(x)^{-1} \Delta x \leq 1\}$. This structural guarantee is encoded as a secondary State-Dependent LMI constraint solved simultaneously:

$$\begin{bmatrix} u_{\max} - \|u_{\text{nom}}(t)\|_2 & Y(x) \\ Y(x)^T & W(x) \end{bmatrix} \succeq 0 \tag{12}$$

7 Simulation Performance Readouts and Trajectory Plots

To verify the performance of the Symmetric Self-Adaptive Contraction Manifold under severe noise constraints, high-fidelity sub-millisecond numerical simulations were executed over a modified IEEE 39-bus topology.

As captured in the exported high-resolution vector trajectory asset (`mpc_contraction_dynamic_response.png`), the grid undergoes an immediate tracking loop failure when an uncoordinated phase noise strike exceeding 2.86° hits the system at time $t = 0.4\,\text{s}$ without intervention. The uncoordinated tracking states break out into a chaotic attractor cascade, overdriving the inner transistor controllers into hard current-limiting saturation.

Conversely, when the in-series preprocessor loop is engaged along the horizontal real-time control path, the Zhou–Köhler MHE parameter estimation network immediately catches the resulting neural residual spike (θ_{NN} shifting from 0.0015 to 0.4856). The adaptive contraction multiplier (α) dynamically morphs the active current constraints inward, compressing the safety limits down to a tighter 0.5420 p.u.. This restriction draws the non-linear trajectories inward, guaranteeing absolute phase-locking convergence and preventing the states from ever hitting the unmodeled current rails up to an extreme stress perimeter of 15.00° of continuous phase noise jitter.

References

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